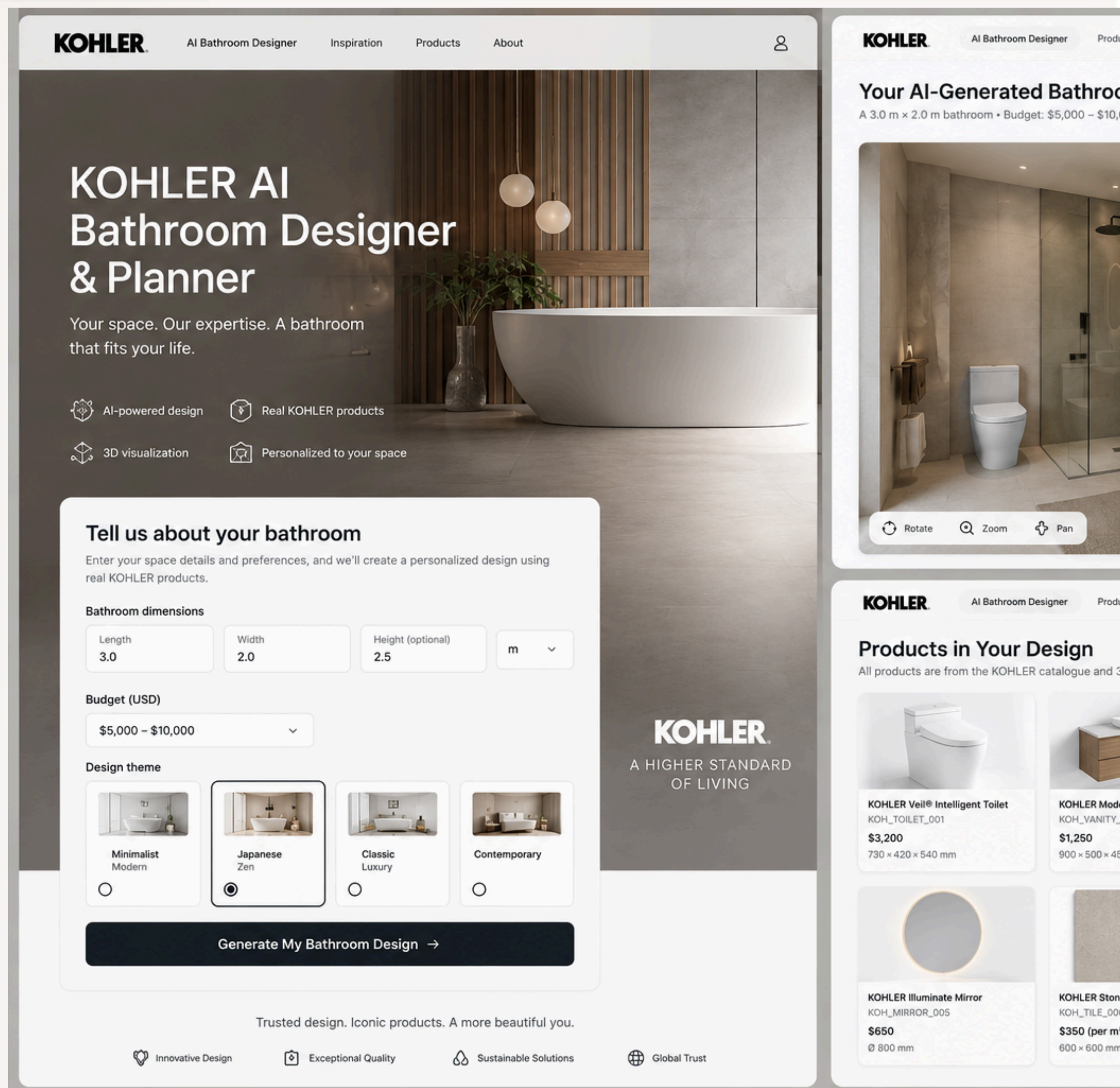


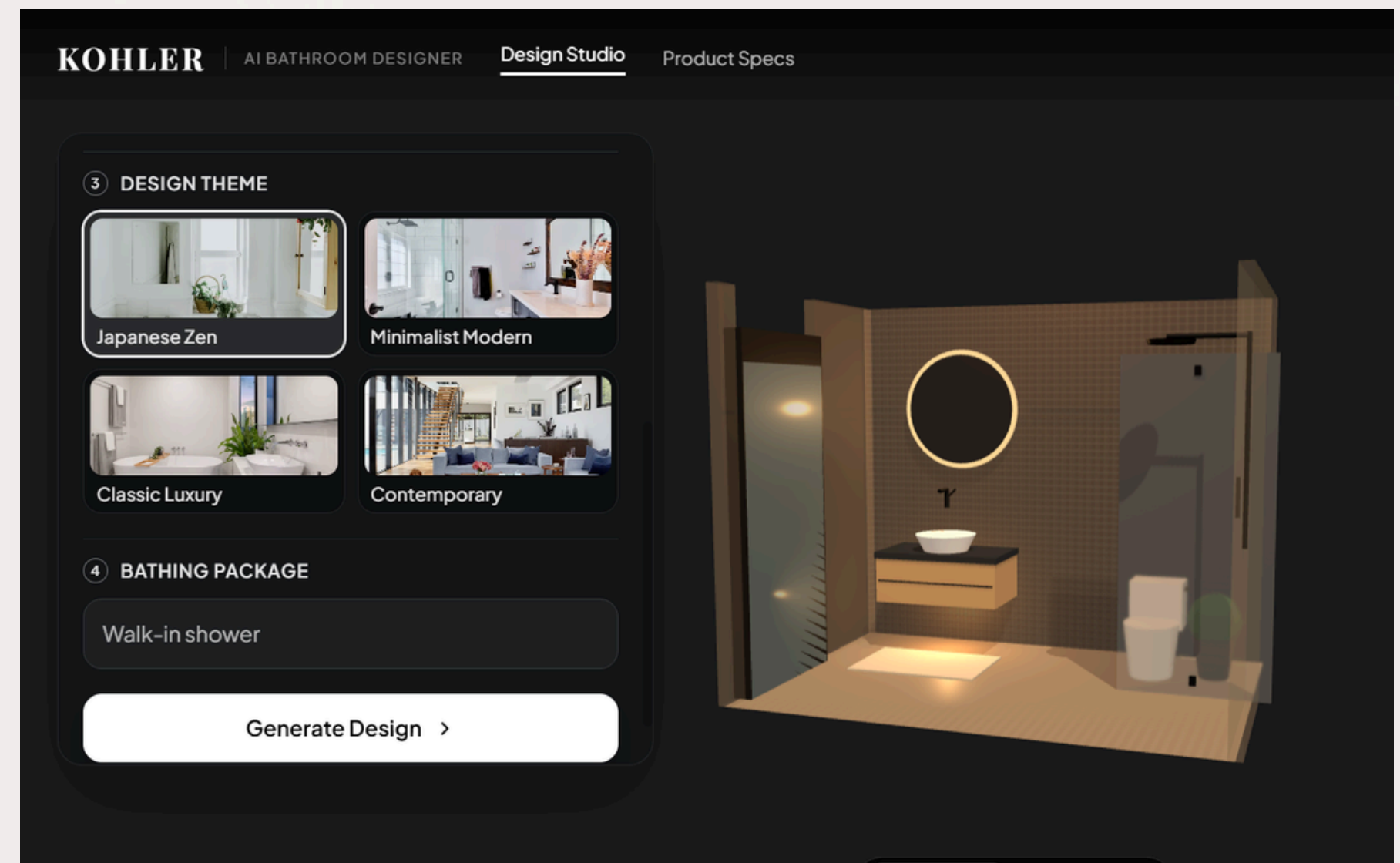
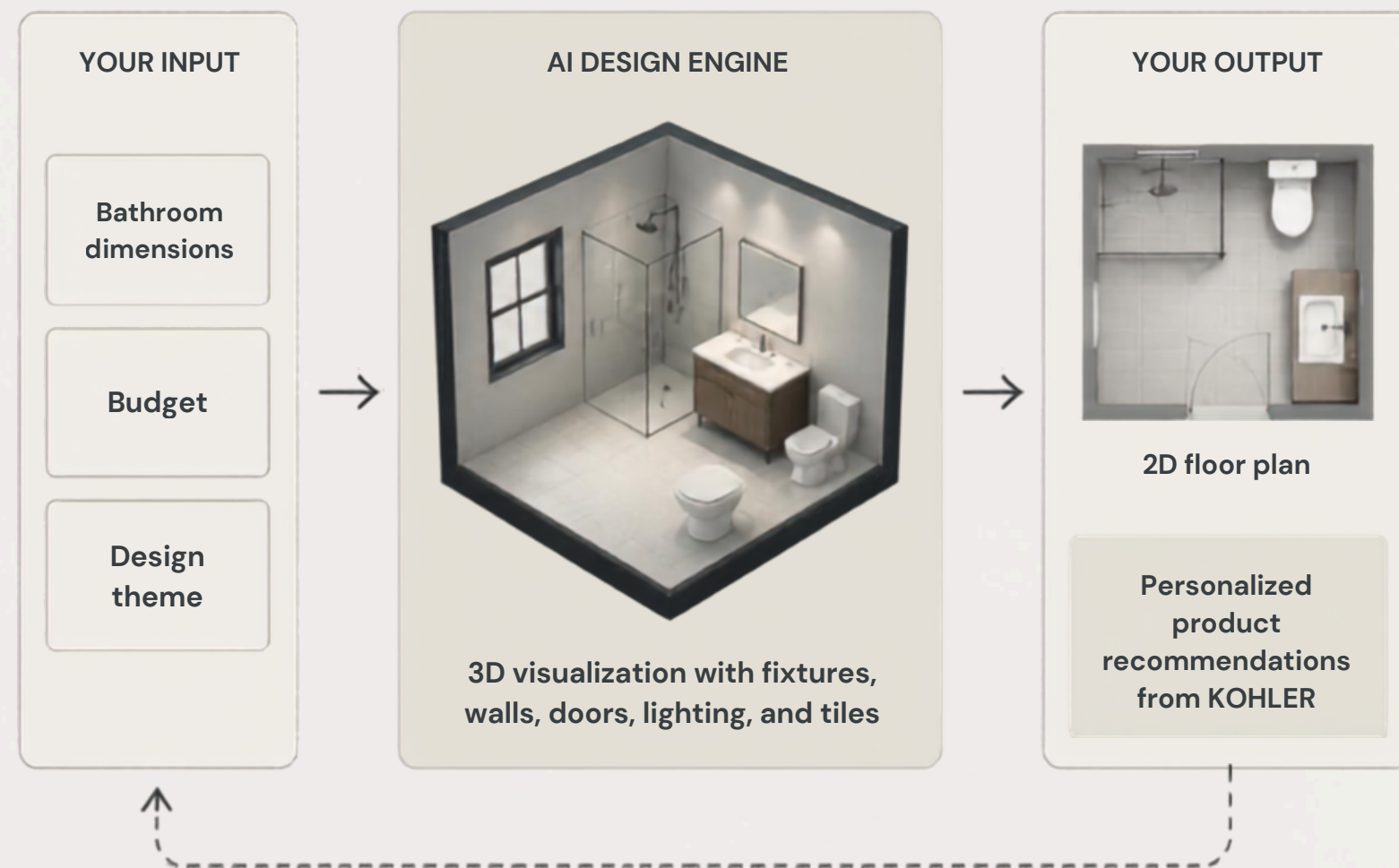
KOHLER AI BATHROOM DESIGNER

Student Prototype for Smart Bathroom
Planning & Recommendations

Kanishka Patil
1262242852
TY BTECH CSE AIDS



AI Bathroom Designer & Planner



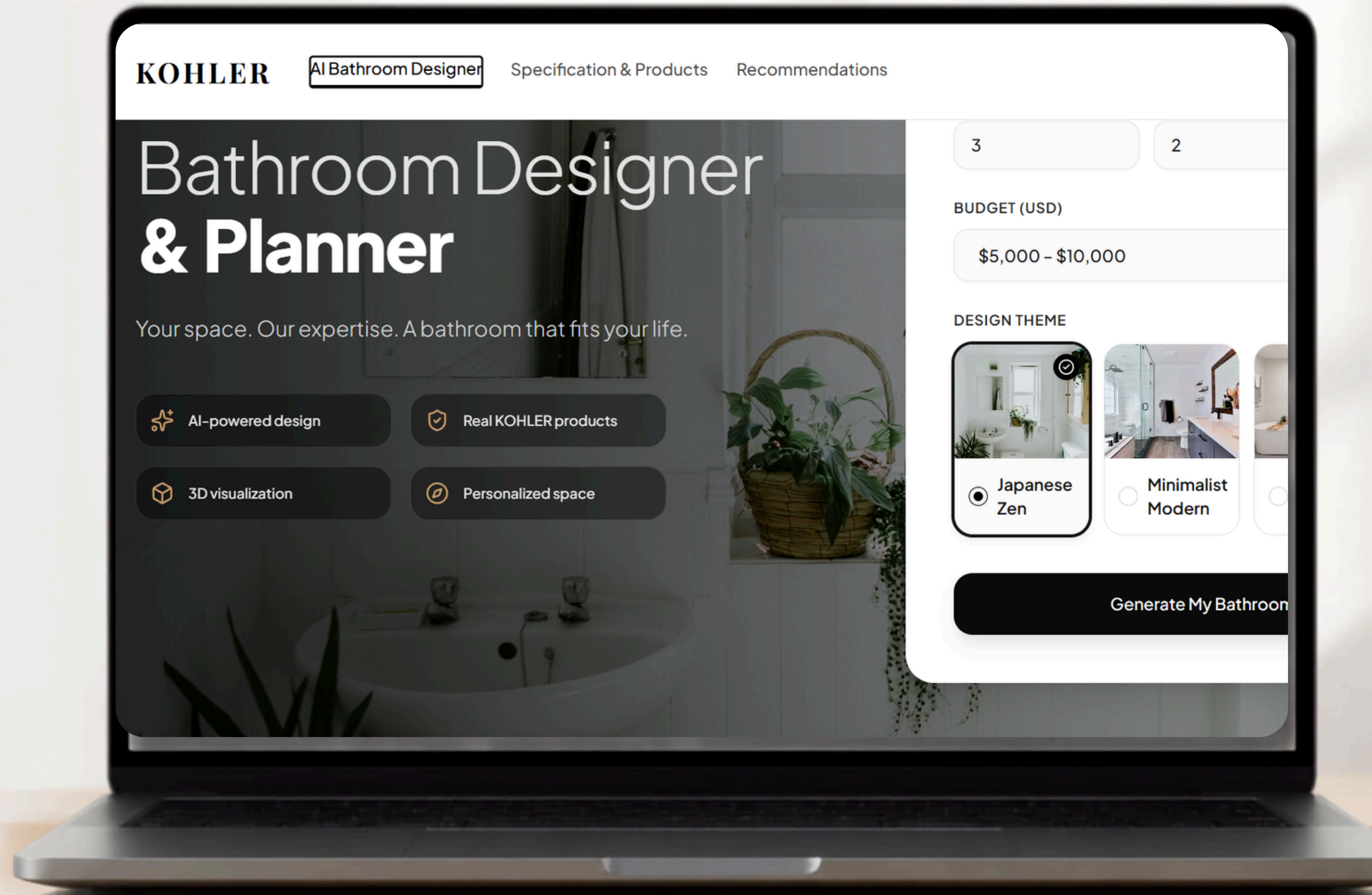
- Catalogue-based KOHLER product recommendations
- Budget and dimension-aware selection
- Theme-aware bathroom configuration
- Interactive 2D/3D visualization with product replacement

Interactive 3D & 2D Visualization Features

Switch seamlessly between
3D and 2D views

Customize themes, tiles,
and finishes in **real time**

Navigate and explore your
bathroom design from
every angle



Tech Stack

Frontend

- React
- TypeScript
- HTML / CSS

3D & Visualization

- Three.js — procedural 3D bathroom and fixtures
- 2D/3D camera views and interactive controls

Build & Development

- Vite
- Node.js 18+
- npm

Application Logic

- TypeScript-based recommendation engine
- Local structured KOHLER product catalogue
- Constraint-based filtering and ranking
- Dimension-aware fixture placement



BUSINESS IMPACT

- Simplifies product discovery and bathroom planning
- Connects customer preferences directly to KOHLER products
- Helps customers visualize combinations before purchase
- Supports budget and space-aware decision making

FUTURE SCOPE

- Live KOHLER catalogue & pricing integration
- Water-efficiency & sustainability scoring
- More accurate product-level 3D models

